

Name:

Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL
(Est. 1911)



Year 11

Mathematics Extension 1
May 2013

Time allowed: 70 min

Instructions:

- Write your name and class at the top of this page.
- These questions must be handed in on the *top* of your answers
- Attempt all questions.
- Begin each question on a new page.

Use only blue or black pen for your answers

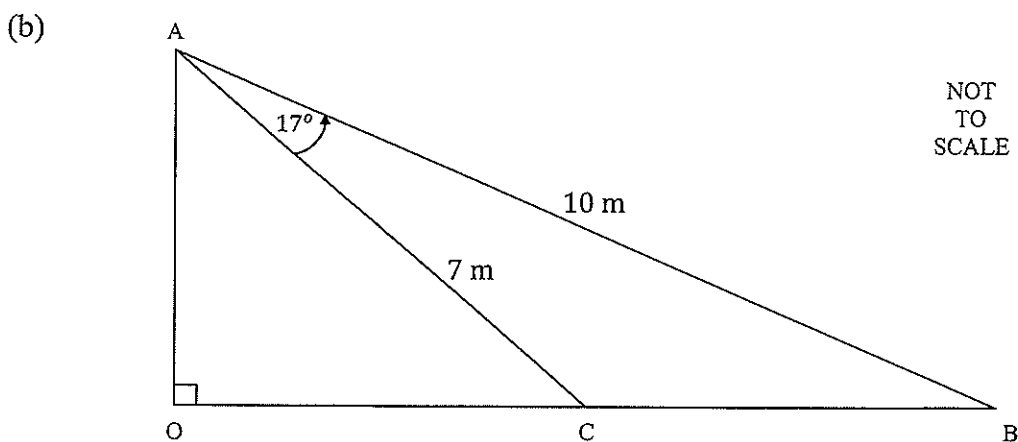
Total Marks – 60

Question 1 (10 marks) Use a SEPARATE page

- (a) Simplify the following expression $\frac{x^3 - y^3}{x^2 - y^2}$. 2
- (b) (i) By factorising, simplify $2^{n+1} + 2^n$. 1
- (ii) Hence, or otherwise, write $\frac{2^{1001} + 2^{1000}}{3}$ as a power of 2. 2
- (c) Simplify: $\frac{10^x + 15^x}{2^5 \times 3^x + 2^{x+5}}$. 3
- (d) Find the exact value of $\sin 120^\circ - \tan 210^\circ$. Express your answer with a rational denominator. 2

Question 2 (10 marks) Use a SEPARATE page.

- (a) The sum of the interior angles of a regular polygon is 3960° .
- (i) How many sides does the polygon have? 1
- (ii) Find the size of each interior angle. 1
- (iii) Hence or otherwise find the size of the exterior angle. 1



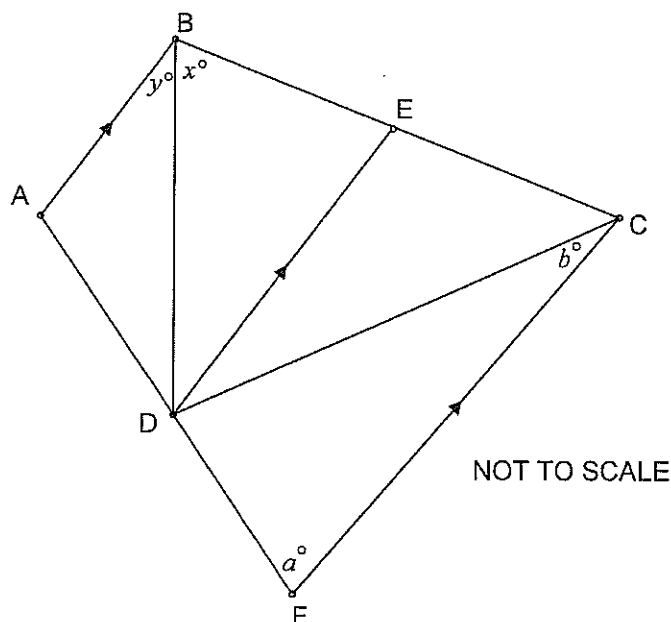
- (i) Find the area of $\triangle ABC$ to 2 significant figures. 2
- (ii) Find the length of BC to 2 significant figures. 2
- (iii) Find the length of BO to 2 significant figures. 3

Question 3 (10 marks) Use a SEPARATE page.

- (a) Find the *exact* solutions of $x + 8 = \frac{6}{x}$ 3
- (b) (i) Draw the graph of $y = |x - 1|$ and $y = x + 3$ on the same axes. 2
- (ii) Hence or otherwise solve $|x - 1| > x + 3$. 2
- (c) Solve for x : 3
- $$\frac{4x - 1}{x + 2} \geq 3$$

Question 4 (10 marks) Use a SEPARATE page.

- (a) In quadrilateral ABCD, $AB = AD$, $CB = CD$ and FC is parallel to AB and DE.



- (i) Show that $a = 2y$, giving reason(s). 3
- (ii) Show that $b = x - y$, giving reason(s). 2
- (b) (i) Use the method of grouping in pairs to factorise fully 2
- $$3x^3 + 3x^2 - x - 1.$$
- (ii) Hence or otherwise solve 3
- $$3\tan^3\theta + 3\tan^2\theta - \tan\theta - 1 = 0 \text{ for } 0 \leq \theta \leq 180^\circ.$$

Question 5 (10 marks) Use a SEPARATE page.

- (a) For the function $f(x) = \frac{9}{9-x^2}$
- (i) Giving reasons, is the function odd, even or neither? 1
 - (ii) Find the equation(s) of the asymptotes. 2
 - (iii) Using a ruler, sketch the graph of $y = f(x)$, showing all key features. 3
 - (iv) Hence, or otherwise, state the domain and range of the function. 1
- (b) If $3 \cos \theta + 2 = 0$ and $\tan \theta > 0$, what is the exact value of $\sin \theta$? 3

Question 6 (10 marks) Use a SEPARATE page.

- (a) Jade is on a ship and observes two lighthouses on the shore. The lighthouse at Addison Head has a bearing of 224° from the ship. The lighthouse at Blake Beach has a bearing of 195° from the ship and 165° from Addison Head. The lighthouses are 3.4 km apart.
- (i) Draw a diagram showing all necessary information. 2
 - (ii) What is the distance of Jade's ship from the Addison Head lighthouse (1 decimal place)? 2
- (b) Prove 3
- $$\frac{(1 + \tan^2 \theta) \cot \theta}{\operatorname{cosec}^2 \theta} = \tan \theta$$
- (c) If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, find the values of a and b . 3

End of test

Q1

$$\begin{aligned}
 (a) & \frac{(x-y)(x^2+xy+y^2)}{(x-y)(x+y)} \\
 &= \frac{x^2+xy+y^2}{(x+y)}
 \end{aligned}$$

$$\begin{aligned}
 (b)(i) & 2 \times 2^n + 2^n \\
 &= 2^n(2+1) \\
 &= \underline{\underline{3 \times 2^n}}
 \end{aligned}$$

$$\begin{aligned}
 (ii) & \frac{2 \times 2^{1000} + 2^{1000}}{3} \\
 &= \frac{3 \times 2^{1000}}{3} = \underline{\underline{2^{1000}}}
 \end{aligned}$$

$$\begin{aligned}
 (c) & \frac{2^x 5^x + 3^x 5^x}{2^5 \times 3^x + 2^5 \times 2^x} \\
 &= \frac{5^x(2^x + 3^x)}{2^5(3^x + 2^x)} \\
 &= \underline{\underline{\frac{5^x}{2^5}}}
 \end{aligned}$$

$$\begin{aligned}
 (d) & \sin(180-60) - \tan(180+30) \\
 &= \sin 60 - \tan 30 \\
 &= \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} \\
 &= \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{3} \\
 &= \frac{3\sqrt{3} - 2\sqrt{3}}{6} = \underline{\underline{\frac{\sqrt{3}}{6}}}
 \end{aligned}$$

Q2

$$\begin{aligned}
 (a) (i) & (3960 \div 180) + 2 = \underline{\underline{24}} \\
 (ii) & 3960 \div 24 = \underline{\underline{165^\circ}} \\
 (iii) & 180 - 165 = \underline{\underline{15^\circ}}
 \end{aligned}$$

$$\begin{aligned}
 (b)(i) & A = \frac{1}{2} \times 10 \times 7 \times \sin 17 \\
 &= \frac{1}{2} \times 10 \times 7 \times 0.2924 \\
 &= \underline{\underline{10 \text{ m}^2 \text{ (2 s.f.)}}}
 \end{aligned}$$

$$\begin{aligned}
 (ii) & BC^2 = 10^2 + 7^2 - 2 \times 10 \times 7 \times \cos 17 \\
 &= 15.117 \dots \\
 \therefore BC &= \underline{\underline{3.9 \text{ m}}}
 \end{aligned}$$

$$(iii) \frac{\sin B}{7} = \frac{\sin 17}{3.9}$$

$$\begin{aligned}
 \sin B &= 0.5247692 \dots \\
 B &= \underline{\underline{31.6527^\circ}}
 \end{aligned}$$

$$\therefore \frac{OB}{10} = \cos 31.6527$$

$$\begin{aligned}
 OB &= 10 \cos 31.6527 \\
 &= \underline{\underline{8.5 \text{ m}}}
 \end{aligned}$$

Q3

$$(a) \quad x^2 + 8x = \frac{6 \times x}{x}$$

$$x^2 + 8x = 6$$

$$x^2 + 8x - 6 = 0$$

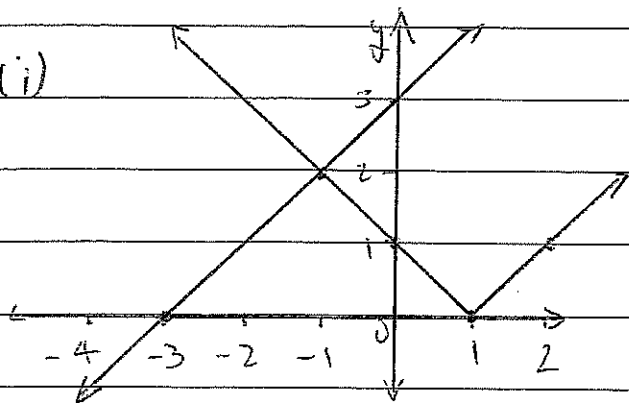
$$x = \frac{-8 \pm \sqrt{8^2 - 4 \times 1 \times (-6)}}{2 \times 1}$$

$$= \frac{-8 \pm \sqrt{88}}{2}$$

$$= \frac{-8 \pm 2\sqrt{22}}{2}$$

$$x = \underline{\underline{-4 \pm \sqrt{22}}}$$

b)(i)

(ii) when $x < -1$

$$c) \quad \frac{4x-1}{x+2} \times (x+2)^2 \geq 3(x+2)^2$$

$$(4x-1)(x+2) \geq 3(x+2)^2$$

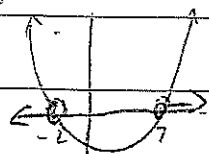
3c) continued.

$$(4x-1)(x+2) - 3(x+2)^2 \geq 0$$

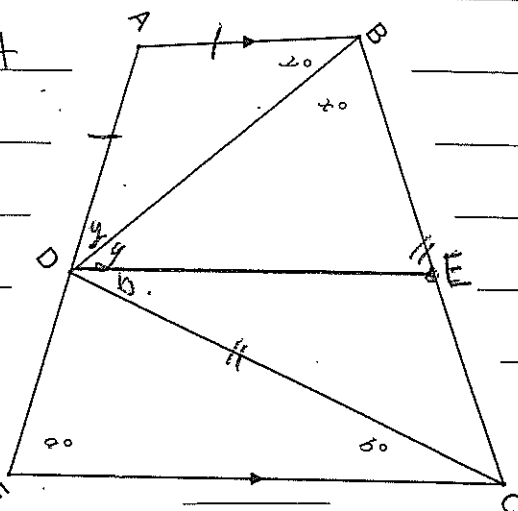
$$(x+2)[4x-1-3(x+2)] \geq 0$$

$$(x+2)(x-7) \geq 0$$

$$\therefore \underline{\underline{x \leq -2 \text{ \& } x \geq 7}}$$



Q4

(i) $AD = AB$ (given)
 $\angle ABD = \angle ADB$ (equal angles of an isosceles $\triangle ABD$)

 $\angle BDE = y$ (alternate angles, $AB \parallel DE$)
 $\therefore a = 2y$ (corresponding angles, $DE \parallel FC$)
(ii) $CB = CD$ (given)
 $\angle CDE = b$ (alternate angles, $DE \parallel FC$)
 $\angle DBC = \angle DCB$ (equal angles of an isosceles $\triangle CBD$)

$$b + y = x$$

$$\therefore b = x - y$$

Q4 continued...

$$(b)(i) 3x^2(x+1) - (x+1)$$

$$(x+1)(3x^2-1)$$

$$\underline{(x+1)(\sqrt{3}x-1)(\sqrt{3}x+1)}$$

$$ii) (\tan\theta+1)(\sqrt{3}\tan\theta-1)(\sqrt{3}\tan\theta+1) = 0 \therefore \sin\theta = -\frac{\sqrt{5}}{3}$$

$$\therefore \tan\theta = -1 \quad \tan\theta = \frac{1}{\sqrt{3}} \quad \tan\theta = -\frac{1}{\sqrt{3}}$$

$$\underline{\theta = 135^\circ}$$

$$\underline{\theta = 30^\circ}$$

$$\underline{\theta = 150^\circ}$$

Q5

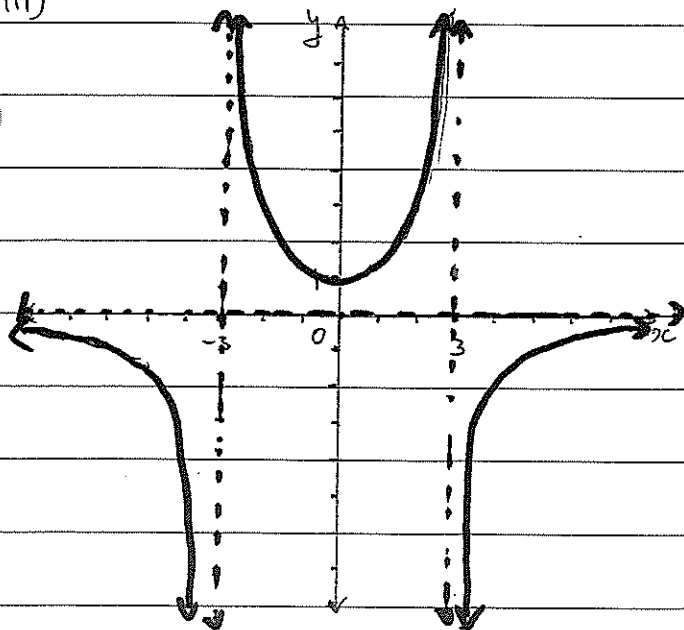
$$(i) f(-x) = \frac{6}{9-(-x)^2} = \frac{6}{9-x^2} = f(x) \therefore \text{even.}$$

$$(ii) 9-x^2 \neq 0$$

$$\therefore \underline{x \neq \pm 3}$$

$$\underline{y \neq 0}$$

(iii)



$$iv) R: \underline{y > 1 \text{ and } y < 0} \text{ D: all } x, \text{ except}$$

$$\underline{x \neq \pm 3}$$

$$(b) \cos\theta = -\frac{2}{3}$$

$$\cos\theta < 0$$

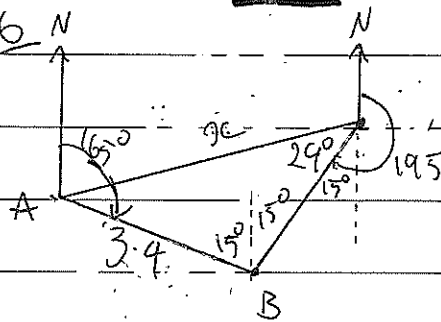
$$\tan\theta > 0$$

$$\therefore \text{3rd quadrant}$$


$$\sqrt{3^2 - (-2)^2} = \sqrt{5}$$

Q6

(i)



$$ii) \frac{x}{\sin 30^\circ} = \frac{3.4}{\sin 29^\circ}$$

$$\therefore \underline{x = 3.5 \text{ km}}$$

$$(b) \sec^2\theta \times \cos\theta = \frac{1}{\sin^2\theta} \times \cos\theta = \frac{1}{\cos\theta} \times \frac{\cos\theta}{\sin\theta} = \frac{1}{\sin\theta} = \frac{1}{\cos\theta} = \tan\theta = \text{RHS}$$

$$(c) \text{Let } x = 2^a \text{ \& } y = 3^b$$

$$x + y = 17 \quad (1)$$

$$2^{a+2} - 3^{b+1} = 5 \Rightarrow 2^2 \times 2^a - 3 \times 3^b = 5$$

$$\therefore 4x - 3y = 5 \quad (2)$$

$$3 \times (1) \quad 3x + 3y = 51 \quad (3)$$

$$(3) + (2) \quad 7x = 56 \therefore x = 8$$

$$8 + y = 17$$

$$2^a = 8$$

$$y = 9$$

$$\therefore \underline{a = 3}$$

$$\therefore 3^b = 9$$

$$\therefore \underline{b = 2}$$